

IMO Training – Number Theory

Selected IMO Shortlist Problems

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Problems

1. (2010 N1) Find the least $n \in \mathbb{N}$ for which there exists a set $\{s_1, s_2, \dots, s_n\}$ consisting of n distinct positive integers such that

$$\left(1 - \frac{1}{s_1}\right) \left(1 - \frac{1}{s_2}\right) \cdots \left(1 - \frac{1}{s_n}\right) = \frac{51}{2010}.$$

2. (2010 N2) Find all pairs (m, n) of nonnegative integers for which

$$m^2 + 2 \cdot 3^n = m(2^{n+1} - 1).$$

3. (2011 N1) For any $d \in \mathbb{N}$, let $f(d)$ be the smallest possible integer that has exactly d positive divisors. (e.g. $f(1) = 1, f(5) = 6, f(6) = 12$) Prove that for every non-negative integer k ,

$$f(2^k) \mid f(2^{k+1}).$$

4. (2012 N1) Call admissible a set A of integers that has the following property: If $x, y \in A$ (possibly $x = y$) then $x^2 + kxy + y^2 \in A$ for every integer k . Determine all pairs m, n of nonzero integers such that the only admissible set containing both m and n is the set of all integers.

5. (2012 N2) Find all triples of positive integers (x, y, z) such that $x \leq y \leq z$ and

$$x^3(y^3 + z^3) = 2012(xyz + 2).$$

6. (2013 N1) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $\forall m, n \in \mathbb{N}$,

$$m^2 + f(n) \mid mf(m) + n.$$

7. (2013 N2, Problem 1) Assume that k and n are two positive integers. Prove that there exist positive integers $m_1, \dots, m_k$ such that

$$1 + \frac{2^k - 1}{n} = \left(1 + \frac{1}{m_1}\right) \cdots \left(1 + \frac{1}{m_k}\right).$$

8. (2014 N2) Find all $x, y \in \mathbb{N}$ such that

$$\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1.$$

9. (2015 N1) Find all $M \in \mathbb{N}$ such that the sequence $(a_n)_{n \geq 0}$ defined by

$$a_0 = M + \frac{1}{2} \text{ and } a_{k+1} = a_k \lfloor a_k \rfloor$$

contains at least one integer term.

10. (2016 N1) For any $k \in \mathbb{N}$, denote the sum of digits of k in base 10 by $S(k)$. Find all polynomials $P(x)$ with integral coefficients such that for any positive integer $n \geq 2016$, $P(n) > 0$ and

$$S(P(n)) = P(S(n)).$$

11. (2016 N3, Problem 4) A set of positive integers is called fragrant if it contains at least two elements and each of its elements has a prime factor in common with at least one of the other elements. Let $P(n) = n^2 + n + 1$. What is the least possible positive integer value of b such that there exists a non-negative integer a for which the set

$$\{P(a+1), P(a+2), \dots, P(a+b)\}$$

is fragrant?

12. (2017 N1, Problem 1) For each integer $a_0 > 1$, define the sequence $(a_n)_{n \geq 0}$ by

$$a_{n+1} = \begin{cases} \sqrt{a_n} & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3 & \text{otherwise.} \end{cases}$$

Find all values of a_0 such that there exists A with $a_n = A$ for infinitely many values of n .

13. (2018 N1) Find all distinct $a, b \in \mathbb{N}$ such that there exists $k \in \mathbb{N}$ for which the number of divisors of ka and that of kb are equal.
14. (2019 N1, Problem 4) Find all pairs (k, n) of positive integers such that

$$k! = (2^n - 1)(2^n - 2)(2^n - 4) \cdots (2^n - 2^{n-1}).$$

15. (2019 N2) Find all $a, b, c \in \mathbb{N}$ such that

$$a^3 + b^3 + c^3 = (abc)^2.$$

16. (2021 N1) Find all positive integers $n \geq 1$ such that there exists a pair (a, b) of positive integers, such that $a^2 + b + 3$ is not divisible by the cube of any prime, and

$$n = \frac{ab + 3b + 8}{a^2 + b + 3}.$$

17. (2021 N2, Problem 1) Let $n \geq 100$ be an integer. Ivan writes the numbers $n, n+1, \dots, 2n$ each on different cards. He then shuffles these $n+1$ cards, and divides them into two piles. Prove that at least one of the piles contains two cards such that the sum of their numbers is a perfect square.
18. (2022 Problem 5) Find all triples (a, b, p) of positive integers with p prime and

$$a^p = b! + p.$$